



BASE24 AXCI ISO Interface Manual

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Version Record

11/08, Version 5, AMEX CAPN EMV Support
03/07, Version 4, Address Verification and Contactless Magnetic
Stripe Support.
05/06, Version 3, AMEX Card Acceptance & Processing Network (CAPN)
Support.
11/05, Version 2, Generic Switch Interface Acquirer Information File
(AIF) Support
05/05, Version 1, AXCI ISO Interface Software Release 6.0

Document History

19MAY2005	owallad
Symptom:	Initial global release.
Problem:	None.
Fix:	Modified the header box to conform to BASE24 programming standards.
Dependency:	Restore release 6.0 files, modify the appropriate CUSTMACS flags, and run MAKE.
Reference:	WO #050111-03
17NOV2005	hayest
Symptom:	The Generic Switch Interface has been modified to support the Acquirer Information File (AIF). It will read the AIF into extended memory and will support the new AIF file LCONF assign.
Problem:	All interfaces that bind with SWCH must have new CUSTMACS flag AIF_ON set to true, regardless of if the interfaces have been modified to utilize the AIF or not, in order for the Generic Switch Interface to successfully compile.
Fix:	Modified section Installation Steps to include the instruction to set CUSTMACS flag AIF_ON to true. Modified section Interface Process Files to include SW60AIF in the list of subvolumes needed for the interface.
Dependency:	Restore new subvolume SW60AIF. Apply fix to CUSTMACS, FLGSDOC, SWIFM, SWIM, SWIMM, SWCHG, SWCHMM and SWCHS. Run MAKE. Replace AXCIMAN.
Reference:	WO #050805-01.
01MAY2006	huac
Symptom:	AMEX Card Acceptance & Processing Network (CAPN) Support.
Problem:	None.
Fix:	Added AMEX Card Acceptance & Processing Network Support.
Dependency:	Apply fixes to COBTKN, DDL PSTKN, PSTKN CVS and PSTKNID. New files: RQAXCIM, RQAXCIS, and SCR NAXCI. Apply fixes to AISODDLS, AXCIDDLM, AXCIDDLS, AXCIG, AXCIMM, AXCIS and SWAXCIM. Replaced AXC ILOGM and AXCIMNWD. Run MAKE.
Reference:	WO #050111-03.
16OCT2006	huac
Symptom:	Multiple Interfaces PCT trace number support.
Problem:	None.
Fix:	Added FIRST-AXCI-TRACE-NUM and INCR-AXCI-TRACE-NUM-BY parameters.
Dependency:	Apply fixes to AXCIDDLS, AXCIG and AXCIS. Run MAKE.
Reference:	Case 421027.

26MAR2007 SWIFTD
Symptom: Address Verification and Contactless Magnetic Stripe Support.
Problem: None.
Fix: Added Support for All Address Verification Formats and Contactless Magnetic Stripe in the AXCI Interface.
Dependency: Apply fixes to AISODDLS, AXCIG, AXCIS, COBTKN, DDL PSTKN, PSTKN CVS and PSTKNID. Replace AXCIMNWD. New file AXCIUPDT. Run Make.
Reference: WO #070207-03 #050209-01

27OCT2008 gruberg
Symptom: AMEX CAPN Interface EMV Support
Problem: None.
Fix: Added support for an EMV bindable module for the AXCI ISO Interface.
Dependency: Apply fixes to CUSTMACS, SWIFM, and SWIMM. Replace FLGSDOC. Apply fixes to AISODDLS, AXCIDDLS, AXCIG, AXCIM, AXCIMM, AXCIS, and SWAXCIM. Replace AXCILOGM, AXCIMNWD and AXCIUPDT. Restore new files AXCID, AXCIEMVM, AXCIEMVS, IAXCFM, IAXCMM, IAXCMNWD and SWIAXCM. Run Make.
Reference: WO #080526-01

Table of Contents

PREFACE	ix
Reading Prerequisites	ix
BASE24 Compatibility	ix
NET24 – XPNET Compatibility	ix
Section 1 INTRODUCTION	1
AXCI ISO INTERFACE PROCESS	1
THE AXCI INTERCHANGE PROCESS	1
THE AXCI ISO INTERFACE PROCESS FUNCTIONS	1
BASE24 FILES	3
CONTROL FILES	3
TRANSACTION FILES	3
LOG MESSAGE DOCUMENTATION	4
Section 2 INSTALLATION AND CONFIGURATION	5
INTRODUCTION	5
AXCI ISO INTERFACE PROCESS FILES	5
INSTALLATION STEPS	8
LCONF ASSIGNS AND PARAMS	13
ASSIGNS	13
PARAMS	16
ICF SETTINGS	21
ICF SCREEN 1 SETTINGS	21
ICF SCREEN 2 SETTINGS	24
ICF SCREEN 3 SETTINGS	26
ICF SCREEN 7 SETTINGS	29
ICF SCREEN 9 SETTINGS	30
ICF SCREEN 10 SETTINGS	30
ICF SCREEN 11 SETTINGS	32
ICF SCREEN 12 SETTINGS	34
KEYF SETTINGS	35
KEYF SCREEN 1 SETTINGS	35
STF SETTINGS	36
TOKEN FILE	39
Section 3 STARTUP AND CONTROL	41
INTRODUCTION	41
PROCESS CONFIGURATION	41
STARTUP	41
EVENT MESSAGES	41
INITIALIZATION	42
NETWORK CONTROL COMMANDS	44
TIMERS	47
OUTBOUND REQUESTS TIMER	47
INBOUND REQUESTS TIMER	47

STORE-AND-FORWARD TIMER	47
CUTOVER TIMER	47
NETWORK MANAGEMENT TIMER	47
EXTENDED NETWORK MANAGEMENT TIMER	48
WAIT FOR TRAFFIC TIMER	48
REPORT STARTUP TIMER	48
PERFORMANCE TIMER	48
Section 4 TRANSACTION HANDLING	51
INTRODUCTION	51
LOGGING TO THE ILF AND SAF	52
TRACE NUMBERS	52
MESSAGE FLOWS FROM BASE24 TO AMEX	53
REQUEST MESSAGE TO AMEX	53
REVERSAL TO AMEX	54
TIMEOUTS OF REQUESTS TO AMEX	55
STORE-AND-FORWARD MESSAGES TO AMEX	56
NETWORK MANAGEMENT MESSAGE FLOWS	57
LOGON/ECHO TEST INITIATED BY BASE24	57
Section 5 CUTOVER AND REPORTING	59
INTRODUCTION	59
CUTOVER PROCESSING	59
REPORTING	59
REPORT OVERVIEW	59
SETL01 CLEARINGS STATEMENT REPORT	60
STAT09 TRANSACTION DETAIL REPORT	60
STARTING REPORTS	60
Appendix A RDT File	61
Introduction	61
CREATING AN RDT FILE	63
OPERATING SNAX/HLS	64
CMI OBEY FILE	65
Appendix B NEFS DECLARATIONS	67
Introduction	67
CMI OBEY FILE	68
SYSGEN LINE CONFIGURATION	68

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PREFACE

The AXCI ISO Interface process is used to establish a communications link between a BASE24 network and the AMEX Credit Interchange. This manual provides basic information for installing, configuring, and operating the AXCI ISO Interface process.

This document is located on the SW60AXCI subvolume in a file called AXCIMNWD. It may be printed using and copied as needed by ACI customers who have purchased ACI's AXCI ISO Interface. The document may also be ordered, at a cost, through ACI Publications.

Reading Prerequisites

While no firm prerequisites exist for reading this manual, it is helpful if readers are somewhat knowledgeable about BASE24 and BASE24 files maintenance.

BASE24 Compatibility

The information contained in this manual is applicable to BASE24 Release 6.0 and above.

NET24 – XPNET Compatibility

The information in this manual is applicable to XPNET 3.0 and above. For previous releases, refer to the appropriate NET24 - XPNET manuals.

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Section 1 INTRODUCTION

AXCI ISO INTERFACE PROCESS

This section introduces the AXCI ISO Interface process and provides an overview of AXCI ISO Interface process functionality. This section also identifies the BASE24 files used by the AXCI ISO Interface process.

THE AXCI INTERCHANGE PROCESS

The AXCI Interchange serves as a link between two networks. It allows a cardholder on one network to initiate a transaction at a terminal belonging to another network.

AMEX connects an acquirer with an issuer. Both the acquirer and the issuer must conform to AMEX's standards. These standards include sending and accepting the appropriate form of the AMEX external message and providing the correct message flows. AMEX standards may be found in the current release of the American Express Global Credit Authorization Guide.

THE AXCI ISO INTERFACE PROCESS FUNCTIONS

The AXCI ISO Interface process is responsible for maintaining the relationship between BASE24 and AMEX. In order to maintain this relationship, the AXCI ISO Interface process must perform a variety of functions. These functions include the following:

- Facilitating message exchange between BASE24 and AMEX.

When BASE24 receives an inbound transaction from AMEX, the AXCI ISO Interface process translates the message into the appropriate internal message and sends the message to a BASE24 Authorization process. When BASE24 receives a transaction and sends it to AMEX, the AXCI ISO Interface process translates the message from the internal message format to the external message format and forwards the message to AMEX.

- Providing store-and-forward capabilities.

When the link between BASE24 and AMEX is down, the AXCI ISO Interface process stores non-interactive messages intended for AMEX in its Store-and-Forward File (SAF). Non-interactive messages are messages not requiring an immediate response to complete a transaction. When the link is restored, the messages are retrieved from the SAF and sent to AMEX.

- Facilitating settlement between BASE24 and AMEX.

The AXCI ISO Interface process processes transactions based on AMEX's reporting day. Since AMEX's cutover rarely occurs at the same time as the BASE24 cutover, the reporting day and the transactions to be settled will differ between the two. The AXCI ISO Interface process produces a set of interchange reports to reconcile these points of view. These reports allow the BASE24 participants to reconcile the settlement positions of AMEX and BASE24.

- Managing communications between BASE24 and AMEX.

The AXCI ISO Interface process sends and receives network management messages to determine the status of the link between BASE24 and AMEX. Network management messages include logon and echo test messages.

BASE24 FILES

The AXCI ISO Interface process uses several files. These files can be divided into control files and transaction files and are identified below.

CONTROL FILES

Control files contain configuration information and parameters that determine how the AXCI ISO Interface process performs certain functions. The control files used by the AXCI ISO Interface process include the following:

- Interchange Configuration File (ICF)
- Key File (KEYF)
- Logical Network Configuration File (LCONF)
- Network Environment Source File (NEFS)
- Token File (TKN)

For more information on these files and how they are used by the AXCI ISO Interface process, see section 2.

TRANSACTION FILES

Transaction files are used to store information about individual transactions processed through the AXCI ISO Interface process. The transaction files used by the AXCI ISO Interface process include the following:

- Interchange Log File (ILF)
- Store-and-Forward File (SAF)

For more information on the ILF and SAF files, see the "BASE24 Files and Record Structures Manual".

LOG MESSAGE DOCUMENTATION

Log messages generated by the AXCI ISO Interface process are documented in the SW60AXCI subvolume in a file named AXCILOGM.

Section 2 INSTALLATION AND CONFIGURATION

INTRODUCTION

This section lists the steps required to install the AXCI ISO Interface process. In addition, it provides guidelines for configuring the required control files, such as the LCONF, ICF, and KEYF.

AXCI ISO INTERFACE PROCESS FILES

The AXCI ISO Interface process can be run in a 3.4/4.0, 5.0, 5.1.2, 5.3 or 6.0 BASE24 environment. The most current release of interface code (i.e. Fortify.3 or Release 6.0) contained in the BASE24 environment dictates what MAKE stream the AXCI ISO Interface process files are made in.

The files needed for the AXCI ISO Interface process are included on five subvolumes: SW60AIF, SW60SWCH, SW60SI92, SW60AXCI and BA60IS92. The following is a list of the files on these subvolumes.

SW60AIF SUBVOLUME

The following is a list of all files on the SW60AIF subvolume.

AIFCOB	- DDL COBOL object for Acquirer Information File
AIFDDL	- MAKE file for Acquirer Information File's DDLs
AIFDDL	- Source file for Acquirer Information File's DDLs
AIFFM	- MAKE configuration file for Acquirer Information File
AIFFUP	- DDL FUP object for Acquirer Information File
AIFMM	- MAKE macro file for Acquirer Information File
AIFTAL	- DDL TAL file for Acquirer Information File
RQAIFM	- MAKE file for Acquirer Information File Requester
RQAIFS	- Source file for Acquirer Information File Requester
SCRNAIF	- Screen source for Acquirer Information File
SVAIF	- COBOL object file for Acquirer Information File Server
SVAIFM	- MAKE file for Acquirer Information File Server
SVAIFS	- Source file for Acquirer Information File Server
SWAIFM	- MAKE file for SW60AIF subvolume

SW60SWCH SUBVOLUME

The following is a list of all files on the SW60SWCH subvolume used by the AXCI ISO Interface.

ISO87DLS	- Source for ISO 87 DDL's
SWCH	- Object file for Generic Switch Interface
SWCHCOB	- DDL COBOL object for Generic Switch Interface
SWCHD	- Dummy proc file for Generic Switch Interface

SWCHDDL - Make file for Generic Switch DDL's
SWCHDDL - DDL source file for Generic Switch Interface
SWCHE - Externals file for Generic Switch Interface
SWCHF - Make configuration file for Generic Switch Interface
SWCHFUP - FUP command source file for Generic Switch Interface
SWCHG - Globals file for Generic Switch Interface
SWCHM - Make file for the Generic Switch Interface
SWCHMM - Make macro file for Generic Switch Interface
SWCHO - Object file for Generic Switch Interface
SWCHS - Source file for Generic Switch Interface
SWCHTAL - DDL TAL file sourced in by SWCHS
SWSWCHM - Make file for SW*SWCH subvolume

The AXCI ISO Interface does not use the following SW60SWCH files:

RQSIIFM - Make file for IIF requester
RQSIIFO - Pseudo target for RQSIIFS
RQSIIFS - Source file for IIF requester
SCRNSIIF - Screen source for Generic Switch-specific functions
SVSIIF - Object file for IIF server
SVSIIFM - Make file for IIF server
SVSIIFS - Source file for IIF server

SW60SI92 SUBVOLUME

The following is a list of all files on the SW60SI92 subvolume.

SI92D - Dummy proc file for SI92 Generic Iso Interface
SI92DDL - Make file for DDL's for the SI92 Generic ISO Interface
SI92DDL - DDL source file for SI92 Generic ISO Interface
SI92E - Externals file sourced in by SI92 Generic ISO Interface
SI92FM - Make configuration for the SI92 Generic ISO Interface
SI92G - Globals source file for SI92 Generic ISO Interface
SI92M - Make file for the SI92 Generic ISO Interface
SI92MM - Make macro file for the SI92 Generic ISO Interface
SI92O - SI92 object used by AXCI ISO Interface bind file
SI92S - Source file for SI92 Generic ISO Interface
SI92TAL - DDL TAL file sourced in by SI92S
SWSI92M - Make file for the SW60SI92 subvolume

* This file will only exist when enhancements or major fixes have been made to the SI92 Generic ISO Interface.

SW60AXCI SUBVOLUME

The following is a list of all files on the SW60AXCI subvolume.

AISODDL - DDL source file for AXCI ISO Interface
AXCI - Executable object for the AXCI ISO Interface
AXCID - Dummy procedures for the AXCI ISO Interface
AXCIDDL - DDL Make file for the AXCI ISO Interface
AXCIDDL - DDL source file for AXCI ISO Interface
AXCIE - Externals file for the AXCI ISO Interface
AXCIFM - Make configuration file for the AXCI ISO Interface
AXCIG - Globals source file for the AXCI ISO Interface
AXCILOGM - Log message file for the AXCI ISO Interface
AXCIM - Make file for the AXCI ISO Interface

AXCIMNWD - Source file of this manual
AXCIMM - Make macro file for the AXCI ISO Interface
AXCIO - Object file for the AXCI ISO Interface
AXCIS - Source file for the AXCI ISO Interface
AXCITAL - DDL TAL file sourced in by AXCIS
AXCIUPDT* - Enhancement update documentation
RQAXCIM - Make file for AXCI requester
RQAXCIS - Source file for AXCI requester
SCRNAXCI - Pathway screens for AXCI requester
SWAXCIM - Make file for the SW60AXCI subvolume

* This file will only exist when enhancements or major fixes have been made to the AXCI ISO Interface.

SW60IAXC SUBVOLUME

The following is a list of all files on the SW60IAXC subvolume.

AXCIEMVE - Externals file for the EMV bindable module
AXCIEMVM - Make file for the EMV bindable module
AXCIEMVO - non executable object from the AXCIEMVS compile
AXCIEMVS - Source file for the EMV bindable module
IAXCFM - Make configuration file for the EMV bindable module
IAXCMNWD - Manual for the EMV bindable module
IAXCMM - Make macro file for the EMV bindable module
SWIAXCM - Make file for the SW60IAXC subvolume

BA60IS92 SUBVOLUME

The following is a list of all files on the BA60IS92 subvolume.

DDLIS092 - DDL Source file
IS92M - ISO92 make file
IS92MM - ISO92 make macro file
IS92TAL - DDL TAL file for ISO 92

INSTALLATION STEPS

The steps required to install the AXCI ISO Interface process are described below and are listed in the order in which they should be performed. Examples follow each step of the procedure to clarify the function the procedure is attempting to perform.

When adding a process, line, or station in a pre-XPNET 3.0 network, the system must be stopped and production is shut down during this time. Therefore, adding processes, lines, and stations to the BASE24 network requires careful planning and should only be performed by authorized personnel.

It may be necessary to refer to the following manuals during the course of these installation procedures. These manuals provide additional information for successfully installing an interchange interface process and its accompanying stations and lines.

- ACI MAKE User Guide (Preliminary Version)
 - BASE24-pos Transaction Processing Manual
 - BASE24 Base Files Maintenance Manual
 - BASE24 Logical Network Configuration File Manual
 - BASE24 Text Command Reference Manual
 - BASE24 Tokens Manual
 - BASE24 Transaction Security Manual
 - NET24-XPNET Network Control Command Reference Manual
 - NET24-XPNET Network Control Operations Guide
 - Tandem's Communication Management Interface Operator Reference Manual
 - Tandem's Subsystem Control Facility (SCF) Reference Manual
 - Tandem's System Management Manual
1. Restore the software tape to the appropriate volume and subvolumes. In order to restore the tape, enter the following commands at the TAOL prompt:

```
RESTORE /OUT $$.#AXCI/ $TAPE, *.*.*.LISTALL,TAPEDATE,  
OPEN,MYID,VOL <$your vol>
```
 2. Pre-XPNET 3.0 networks must configure a Tandem port and line for the AXCI station(s) via SYSGEN. If a Tandem communication subsystem controller (CSS 6100) is used, the line should be configured using the EBCDIC character set qualifier.
 3. Volume to \$vol.BA*MAKE and edit CUSTMACS.
 - a) Set CLASSIC_ON to true.
 - b) Set SWITCH_ON to true.
 - c) Set ISO_92_ON to true.
 - d) Set SDF_ON to true.
 - e) Set AMEX_CREDIT_ISO_ON to true.
 - f) Set GENERIC_SWITCH_ON to true.

- g) Set SI92_ON to true.
- h) Set AIF_ON to true.
- i) Set AXCI_EMV_ON if installing the AMEX CAPN EMV extension
- j) If the AXCI ISO Interface will be run in a release 6.0 network environment, set SW_PRE_60_LCONF_ON to false. If the AXCI ISO Interface will be run in a pre-release 6.0 network environment, set SW_PRE_60_LCONF_ON to true.

NOTE: It is recommended that you use the B24C/B24M files so all necessary modules are rebuilt. If using the B24C/B24M files, part b in steps 4 thru 9 may be skipped.

4. Volume to \$vol.SW60SDF and perform the following:

- a) Edit the macro Make file SDFMM and ensure that the system, volume and subvolume references are correct.
- b) Make the SW60SDF subvolume by entering the following command at the TACL prompt:

```
RUN $vol.subvol.makevXXX/ out $S.#SDF/SDFDDL -FORCE
where: XXX is the Make version.
```

5. Volume to \$vol.SW60AIF and perform the following:

- a) Edit the macro Make file AIFMM and ensure that the system, volume and subvolume references are correct.
- b) Make the SW60AIF subvolume by entering the following command at the TACL prompt:

```
RUN $vol.subvol.makevXXX/ out $S.#AIF/SWAIFM -FORCE
where: XXX is the Make version.
```

6. Volume to \$vol.SW60SWCH and perform the following:

- a) Edit the macro Make file SWCHMM and ensure that the system, volume and subvolume references are correct.
- b) Make the SW60SWCH subvolume by entering the following command at the TACL prompt:

```
RUN $vol.subvol.makevXXX/ out $S.#SWCH/SWSWCHM -FORCE
where: XXX is the Make version.
```

7. Volume to \$vol.BA60IS92 and perform the following:

- a) Edit the macro Make file IS92MM and ensure that the system, volume and subvolume references are correct.
- b) Make the BA60IS92 subvolume by entering the following command at the TACL prompt:

```
RUN $vol.subvol.makevXXX/out $S.#ISO92/IS92M -FORCE
where: XXX is the Make version.
```

8. Volume to \$vol.SW60SI92 and perform the following:

- a) Edit the macro Make file SI92MM and ensure that the system, volume and subvolume references are correct.
- b) Make the SW60SI92 subvolume by entering the following command

at the TACL prompt:

```
RUN $vol.subvol.makevXXX/out $S.#SI92/SWSI92M -FORCE
where: XXX is the Make version.
```

9. Volume to \$vol.SW60IAXC and perform the following:

- a) Edit the macro Make file IAXCMM and ensure that the system, volume and subvolume references are correct.
- b) Make the SW60IAXC subvolume by entering the following command at the TACL prompt:

```
RUN $vol.subvol.makevXXX/out $S.#AXCI/SWIAXCM -FORCE
where: XXX is the Make version.
```

10. Volume to \$vol.SW60AXCI and perform the following:

- a) Edit the macro Make file AXCI MM and ensure that the system, volume and subvolume references are correct.
- b) Make the SW60AXCI subvolume by entering the following command at the TACL prompt:

```
RUN $vol.subvol.makevXXX/out $S.#AXCI/SWAXCIM -FORCE
where: XXX is the Make version.
```

11. If part b in steps 4 thru 10 was skipped in order to run the B24C/B24M files, perform the following:

- a) Volume to \$vol.SW60SRC.
- b) Edit Make macro file SWIFM and ensure that the system, volume and subvolume references are correct.
- c) Volume to \$vol.BA60SRC.
- d) Edit B24C and ensure that the -force option is NOT included in the MAKE command unless this is a first time install.
- e) Make all necessary modules on your BASE24 system by entering the following command at the TACL prompt:

```
/IN B24C/
```

12. Make will move the object files to the location specified in the configuration Make file if the move_obj_subvol_on define is set to TRUE. Whoever runs the Make must have security to write to the subvolume where Make attempts to place the objects.

13. Volume to the location of the template files used by the network. Create the ILF and SAF template files by entering the following commands:

```
FUP/IN $vol.BA60SRC.BATPLT, OUT $S.#BATPLT/
FUP/IN $vol.SW60SWCH.SWCHFUP, OUT $S.#SWCHFUP/
```

14. Check the Network Environment Source File (NEFS) and ensure that the following are defined:

- a) A AXCI ISO Interface process declaration
- b) A AXCI ISO station declaration for acquirer and/or issuer processing

- c) A AXCI ISO device declaration
When using a Tandem communication subsystem controller (CSS 6100) the TRANSMIT TRANSPARENT ENABLED declaration must be set
- d) A AXCI ISO line declaration

For details on the previous declarations, refer to the "NET24-XPNET Network Control Command Reference Manual" and Appendix A of this manual.

15. Add the necessary assigns and params to the Logical Network Configuration File (LCONF) for the AXCI ISO Interface. The required assigns and params are documented in the LCONF assigns and params section of this document.

In addition to these assigns and params, also add the symbolic process name of the AXCI ISO Interface as one of the "setl-notify" params and as one of the "warm" params if this process is to be warmbooted with a DCT WARMBOOT ALL command.

Steps 16 thru 18 are not necessary if using XPNET 3.0.
Steps 16 thru 18 are not necessary if using XPNET 3.0.

16. Run the NETGEN program to produce a new NEF object file. Details for this step can be found in the "BASE24 Network Environment Source File Manual". This step is required only if the NEFS has changed.
17. Add any required PRADD statements to the PRECMD file. The PRECMD file contains commands used by the PREGEN program to establish message routing. For information on routing and the PRECMD file, see the "BASE24 Configuration Manual".
18. Perform a SPANGEN by running the SPAN configuration generation programs. This is required only if PRADD statements have been added or the NEFS has changed. The following programs should be run in this order: TOPGEN, PREGEN, VALIDATE, and DISTRIB. Details on the SPAN configuration generation programs can be found in the "BASE24 Configuration Manual".
19. Add an ICF record for the AXCI ISO Interface process using the information provided later in this document and in the "BASE24 Base Files Maintenance Manual".
20. Add a KEYF record for the AXCI ISO Interface process using the information provided later in this document, in the "BASE24 Base Files Maintenance Manual" and in the "BASE24 Transaction Security Manual".

The AXCI ISO Interface process requires a KEYF record but it is not used and can be set to default values.
21. Add the necessary STF records for the AXCI ISO Interface process using the information provided later in this document and in the "BASE24 Base Files Maintenance Manual".

NOTE: The next two steps are not required if you are using XPNET 3.0 or above.

WARNING: The following steps should be performed only by authorized personnel. Shutting down the BASE24 network results in shutting down production. This requires careful planning and should be performed only by authorized personnel.

22. Bring down the BASE24 network, if it is running, by entering the STOP command two consecutive times from a network control facility.
23. Bring the BASE24 network back up by entering the START command from a network control facility.

LCONF ASSIGNS AND PARAMS

The Logical Network Configuration File (LCONF) assigns and params required by the AXCI ISO Interface process are described below.

Refer to the "BASE24 Logical Network Configuration File (LCONF) Manual" for instructions on adding them to the LCONF.

NOTE: Some of these assigns and params are used throughout BASE24. Therefore, they may already be defined in the LCONF.

ASSIGNS

The following assigns are used by the AXCI ISO Interface process. The assigns marked with asterisks (**) are required if reports are to be generated.

ICF

The fully-qualified file name of the Interchange Configuration File. This assign is required.

ILF

The fully-qualified file name of the Interchange Log File. This assign is required.

Each interface process must have its own ILF assign defined in the LCONF. ILF assigns are tied to interface processes by placing the symbolic name of the interface process in the PROCESS NAME field of the LCONF assign screen.

ILF assigns also require that a fully-qualified template file name be specified in the TEMPLATE field of the LCONF assign screen. BASE24 uses the specified template file to create ILFs.

KEYF

The fully-qualified file name of the Key File. This assign is required if the BASE24-REL param is present and contains the value of 50 or 60. In a Release 5.0 or above system this assign is used by the interface to identify the Key File.

KEY5

The fully-qualified file name of the Key File. This assign is required if the BASE24-REL param is not present or does not contain the value of 50 or 60. In a Release 3.4/4.0 system this assign is used by the interface to identify the Key File.

LMCF

The fully-qualified file name of the Log Message Configuration File. This assign is required.

SAF

The physical file name of the Store-and-Forward File. This assign is required.

Each interface process must have its own SAF assign defined in the LCONF. SAF assigns are tied to interface processes by placing the symbolic name of the interface process in the PROCESS NAME field of the LCONF assign screen.

SAF assigns also require that a fully-qualified template file name be specified in the TEMPLATE field of the LCONF assign screen. BASE24 uses the specified template file to create SAFs.

STF

The fully-qualified file name of the Switch Terminal File. This assign is required for acquirer processing.

TKN

The fully-qualified file name of the Token File. This assign is optional and is only used if the BASE24-REL param is present and contains the value of 50 or 60. This assign is only used in a Release 5.0 or above system.

RPT **

The fully-qualified file name of the Report Perusal File used by BASE24.

RPT assigns also require that a fully-qualified template file name be specified in the TEMPLATE field of the LCONF assign screen. BASE24 uses the specified template file to create Report Perusal Files.

RPT-ERROR-FILE **

The spooler location for log messages generated by the Interchange Interface report programs.

SW-RPT-OBEY-FILE **

The fully-qualified file name of the obey file used to start the interchange reports. The interface process creates its report

obey files using this name.

Each interface process must have its own SW-RPT-OBEY-FILE assign defined in the LCONF. SW-RPT-OBEY-FILE assigns are tied to interface processes by placing the symbolic name of the interface process in the PROCESS NAME field of the LCONF assign screen.

SW-RPT01-PROG-NAME **

The fully-qualified file name of the executable object code that generates the interface's SETL01 report program.

The AXCI ISO Interface does not have its own RPT01 report program and uses the standard BASE24 interface report. Therefore, each AXCI ISO Interface process can use the generic SW-RPT01-PROG-NAME assign defined in the LCONF. The AXCI ISO Interface does not need the symbolic name of the interface process in the PROCESS NAME field of the LCONF assign screen.

SW-RPT09-PROG-NAME **

The fully-qualified file name of the executable object code that generates the interface's STAT09 report program.

The AXCI ISO Interface does not have its own RPT09 report program and uses the standard BASE24 interface report. Therefore, each AMXI ISO Interface process can use the generic SW-RPT09-PROG-NAME assign defined in the LCONF. The AXCI ISO Interface does not need the symbolic name of the interface process in the PROCESS NAME field of the LCONF assign screen.

PARAMS

The following params are used by the AXCI ISO Interface process. The params marked with asterisks (**) are required if reports are to be generated.

ACCT-NUM-TYP

A value (B or D) indicating whether account numbers are stored in binary (B) or decimal (D) format. Currently, only decimal format is supported, allowing for account numbers of up to 19 digits in length.

BASE24-REL

The Release of the BASE24 system in which the interface is running. Valid values are:

40 = Release 3.4/4.0
50 = Release 5.0/5.1.2/5.3
60 = Release 6.0 and above (default)

COUNTER-UPDATE-INTERVAL

A value indicating the number of transactions processed before updating the ILF trace record. Because the interchange prefers the trace number in the external messages to be unique, the AXCI ISO Interface process uses the trace record (first record in the ILF) to store the trace number at designated intervals. Valid values are:

0 - 32767 (default 500)

DATE-DISPLAY-TYPE

A value indicating the format of the month and day portion of the dates on reports. When dates include the year, this param does not affect the year portion of the date. Valid values are:

1 = DDMM
2 = MMDD (default)

FIRST-AXCI-TRACE-NUM

The starting value for the trace number. This param is necessary when multiple copies of the AXCI Interface are processing transactions. When used, the value of this param must be unique for each AXCI Interface.

ILF-LOGGING-ENABLED

A flag indicating whether ILF logging is to take place. This param must be set to "Y" to process reversals out to the interchange. Valid values are:

N = ILF logging will not occur
Y = ILF logging will occur (default)

ILF-NOTIFY-INTERVAL

The number of writes made to the Interchange Log File (ILF) between calculating the percentage of file full. Each time a write is made to the ILF, a counter is incremented. When the counter is equal to the value in this param, a calculation is performed to determine how full the ILF is. The counter is then reset to zero. When the file reaches the percent full specified in the ILF-NOTIFY-PERCENTAGE param, a message is logged indicating the file is becoming full. Until more file space is made available, the interface process continues to log messages that the file is becoming full at the interval specified in this param. Valid values are 10 through 32766.

ILF-NOTIFY-PERCENTAGE

The percentage of used file space at which the interface process logs a message indicating the ILF is becoming full. The interface process continues to log messages at the interval indicated in the ILF-NOTIFY-INTERVAL param until more file space is made available. Valid values are 10 through 99.

INCR-AXCI-TRACE-NUM-BY

The value to increment the trace number by. This param is necessary when multiple copies of the AXCI Interface are processing transactions. When used, the value of this param should equal the number of AXCI Interfaces that are being used.

SEND-ETX

This param will let the interface know whether or not to send the end_of_text (ETX) character. This param must be set to N for SNA and other protocols that do not support the ETX character. Valid values are as follows:

Y = Send the ETX. (default)
N = Do not send the ETX.

MAX-STF-RECORDS

The maximum number of records belonging to the AXCI ISO Interface in the Switch Terminal File (STF). This parameter is used to

determine the amount of extended memory required by the interface. If this parameter is set to 0, enough extended memory for all records in the STF is allocated. Valid values are 0 through 32766 (default 0).

POS-LN-CUTOVER

The time (HH:MM) at which all BASE24-pos processes in the logical network are cut over. All institutions must be cut over by this time. Valid values, in military time, are from 00:00 to 23:59 (no default value). This param is required if processing BASE24-pos transactions.

REVERSE-BAL-INQ

A flag indicating whether balance inquiry reversals will be sent into BASE24. Valid values are:

N = Balance inquiry reversals will not be sent into BASE24
Y = Balance inquiry reversals will be sent into BASE24

SAF-NOTIFY-INTERVAL

The number of writes made to the Store-and-Forward File (SAF) between calculating the percentage of file full. Each time a write is made to the SAF, a counter is incremented. When the counter is equal to the value in this param, a calculation is performed to determine how full the SAF is. The counter is then reset to zero. When the file reaches the percent full specified in the SAF-NOTIFY-PERCENTAGE param, a message is logged indicating the file is becoming full. Until more file space is made available, the interface process continues to log messages that the file is becoming full at the interval specified in this param. Valid values are 10 through 32766 (default 50).

SAF-NOTIFY-PERCENTAGE

The percentage of used file space at which the interface process logs a message indicating the SAF is becoming full. The interface process continues to log messages at the interval indicated in the SAF-NOTIFY-INTERVAL param until more file space is made available. Valid values are 10 through 99 (default 70).

SPAN-FLAG

A value indicating whether totals are to be written to the Span Clearings File by the interface process. Valid values are:

Y = Yes, write totals to the Span Clearings File.
N = No, do not write totals to the Span Clearings File (default).

STATION-TYP

This LCONF param is used by the AXCI ISO Interface to determine whether to use station 1 from the ICF as a second acquirer station. Valid values are as follows:

0 = Normal Processing; The interface uses the station(s) from the ICF with station 1 (if present) and station 2 as acquirer station(s).

The default for this param is 0. If a process specific or generic param is not found the interface will still process in this fashion. If a generic station-typ param is found and the value is anything other than zero, the AXCI ISO Interface will not process correctly. Therefore, it is advisable to enter this param in the LCONF for the name of the AXCI ISO Interface process as defined in the Network Environment Source File (NEFS).

SW-ILF-FILE-RETENTION

The number of days the ILF is to remain on the system. After the specified number of days, the ILF is purged from the system by the interface process. Valid values are 3 through 99 (default 7).

SWI-ACQ-AVS-DATA-SUPPRS

A value indicating whether Address Verification (AVS) data will be suppressed before logging to the ILF.
Valid values are:

Y = Suppress AVS data before logging
N = Do not suppress AVS data (default)

SWI-ACQ-CVD2-DATA-SUPPRS

A value indicating whether Card Verification (CVD2) data will be suppressed before logging to the ILF.
Valid values are:

Y = Suppress CVD2 data before logging
N = Do not suppress CVD2 data (default)

SWI-ACQ-DISCR-DATA-SUPPRS

A value indicating whether discretionary data from track 1 and/or will be suppressed before logging to the ILF. Valid values are:

Y = Suppress discretionary data before logging
N = Do not suppress discretionary data (default)

SWI-ISS-AVS-DATA-SUPPRS

A value indicating whether Address Verification (AVS) data will be suppressed before logging to the ILF.

Valid values are:

- Y = Suppress AVS data before logging
- N = Do not suppress AVS data (default)

SWI-ISS-CVD2-DATA-SUPPRS

A value indicating whether Card Verification (CVD2) data will be suppressed before logging to the ILF.

Valid values are:

- Y = Suppress CVD2 data before logging
- N = Do not suppress CVD2 data (default)

SWI-ISS-DISCR-DATA-SUPPRS

A value indicating whether discretionary data from track 1 and/or will be suppressed before logging to the ILF. Valid values are:

- Y = Suppress discretionary data before logging
- N = Do not suppress discretionary data (default)

SW-RPT-STARTUP **

The time (HH:MM) at which the interface process is to run its report obey file. Valid values, in military time, are from 00:00 to 23:59 (no default value).

ICF SETTINGS

Each interface process must have its own Interchange Configuration File (ICF) record. The following describes how the ICF must be configured for the AXCI ISO Interface.

None of the ICF screens are depicted; however, the fields are described as they actually appear on the ICF screens. For actual screen depictions, refer to the "BASE24 Base Files Maintenance Manual".

ICF SCREEN 1 SETTINGS

The following fields on ICF screen 1 must be set as indicated.

INTERCHANGE FIID

The FIID uniquely identifying the AXCI ISO Interchange. This field must match the logical network ID given to the AXCI ISO Interchange in the BASE24 system (from the LOGICAL NETWORK IS element in the Network Environment Source File). This field and the PROCESS field combine to form the primary key to the ICF.

FIELD LENGTH:	1-4 alphanumeric characters
REQUIRED FIELD:	Yes
DEFAULT VALUE:	None

PROCESS

The name of the AXCI ISO Interface process as defined in the Network Environment Source File (NEFS). This is the process that receives and sends transaction messages from and to AMEX. This field and the INTERCHANGE FIID field combine to form the primary key to the ICF.

FIELD LENGTH:	1-16 alphanumeric characters
REQUIRED FIELD:	Yes
DEFAULT VALUE:	None

SWITCH TYPE

A code identifying the interchange. For the AXCI ISO Interface process, this value must be "AXCI".

FIELD LENGTH:	1-4 alphanumeric characters
REQUIRED FIELD:	Yes
DEFAULT VALUE:	None

INTERCHANGE LOGICAL NET

The logical network identifier given to the interchange. This value must match the value in the INTERCHANGE FIID field.

FIELD LENGTH: 1-4 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

REPORTING NAME

The name of the interchange as it is to appear on the BASE24 reports.

FIELD LENGTH: 1-20 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

INSTITUTION ID

The institution identifier assigned by the interchange. Used by the AXCI ISO Interface process to fill field 42 in the external message if a match cannot be found in the STF.

FIELD LENGTH: 1-15 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

SWITCH ID

The identification of the interchange center. Not used by the AXCI ISO Interface process.

FIELD LENGTH: 1-15 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

STATION 1

For normal processing, station 1 is used to send and receive messages to and from interchange for issuer processing. Since the AXCI ISO Interface cannot perform issuer processing, this field can be used to assign a second acquirer station. To use this field as a second acquirer station, the ISSUER option on screen 3 must be set to "N", the station-tyt param in the LCONF must be 0 (normal processing) and the symbolic name must not be the same as the symbolic name used for station 2. This name must match the name of a station added in the NEFS for the AXCI ISO Interface process.

FIELD LENGTH: 1-16 alphanumeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: None

STATION 2

The symbolic name of a AMEX station that sends and receives messages to and from the AXCI ISO Interface process for acquirer processing. This name must match the name of a station added in the NEFS for the AXCI ISO Interface process.

FIELD LENGTH: 1-16 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

SIC CODE

The Standard Industrial Classification (SIC) code applicable to the institution operating the BASE24-atm or BASE24-pos network. This field is not used by the AXCI ISO Interface process.

FIELD LENGTH: 1-4 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: 0

CURRENCY CODE

The acquirer's currency code for transactions sent to AMEX. Valid values are listed in "Codes for the Representation of Currencies and Funds," ISO 4217-1981. For the AXCI ISO Interface process, this field should be set to "840".

FIELD LENGTH: 3 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 840 (US dollars)

DEFAULT TERM NUM

The default terminal number used to fill the external pseudo terminal number field on outbound transactions when an STF record for the terminal does not exist.

FIELD LENGTH: 1-16 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

DEFAULT ACQUIRER ID NUM

A default acquirer ID number used if an inbound transaction from the interchange does not contain an acquirer's institution identification number. Not used by the AXCI ISO Interface process.

FIELD LENGTH: 1-11 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: 00000000000

CUSTOMER BALANCE DISPLAY

AMEX's preference concerning balance presentation for balance inquiries when BASE24 is the acquirer. Valid values are:

0 = Do not display or print (default)
1 = Display only
2 = Print only
3 = Display and print

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: 0 (do not display or print)

ICF SCREEN 2 SETTINGS

The following fields on ICF screen 2 must be set as indicated.

SETTLEMENT INFORMATION

The following fields contain interchange settlement information.

SETTLEMENT HOUR

The hour (local time) at which settlement occurs for the interchange. Valid entries include 00 through 23.

FIELD LENGTH: 2 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: 00

SETTLEMENT MINUTE

The minute (local time) at which settlement occurs for the interchange. Valid entries include 00 through 59.

FIELD LENGTH: 2 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: 00

SETTLEMENT DAYS

A value indicating whether the interchange processes five or seven days a week. Valid values are:

0 = Processes 5 days a week
1 = Processes 7 days a week (default)

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: 1 (process 7 days). If this field is
replaced with a blank, the default changes to
0 (process 5 days).

REPORT PRIORITY

The priority at which report programs are run when started. The range is 0 through 255, with 255 being the highest priority (default 100).

FIELD LENGTH: 1-3 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 100

REPORT CPU

The CPU in which the report programs are started.

FIELD LENGTH: 1-2 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0

SWITCH POSTING DATE (YYMMDD)

The posting date (YYMMDD) for AMEX. This date is updated daily at settlement time and upon initialization of the AXCI ISO Interface process.

FIELD LENGTH: 6 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

HOLIDAY DATES (YYMMDD)

AMEX's defined holiday dates (YYMMDD). AMEX does not settle on these holidays and the AXCI ISO Interface process does not create an Interchange Log File (ILF) for these dates. This feature is only used when the interface is configured for five-day settlement.

FIELD LENGTH: 16 fields of 6 numeric characters each
REQUIRED FIELD: No
DEFAULT VALUE: None

ICF SCREEN 3 SETTINGS

The following fields on ICF screen 3 must be set as indicated.

TIMER LIMITS

The following timer value fields control specific transaction message activity between the BASE24 network and AMEX. The values entered in these fields must be checked against AMEX's timer values to ensure the integrity of message traffic.

NETWORK MANAGEMENT

The number of seconds BASE24 waits before setting an EXTENDED NETWORK timer for an outstanding network management message (i.e., logon, logoff, echo test, key change). Valid values are 0-9999 seconds.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 30

EXTENDED NETWORK

The number of seconds BASE24 waits before attempting to retry a network management message. This timer is used by the interface process to determine the number of seconds to wait before marking the line back up after receiving notification of a failed message. Valid values are 0-9999 seconds.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 60

WAIT FOR TRAFFIC

The number of seconds BASE24 waits for traffic on the line from the interchange before initiating a network management echo test message. Valid values are 0-9999 seconds.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 60

PERFORMANCE PERIOD

The number of minutes the AXCI ISO Interface process uses as an interval for tracking performance of itself. Valid values are 0-9999.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 20

MAXIMUM OUTSTANDING TRANSACTIONS

The following fields specify the maximum number of outstanding inbound and outbound transactions.

OUTBOUND

The maximum number of outstanding outbound transactions. Valid values are 0-9999 transactions.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 15

INBOUND

The maximum number of outstanding inbound transactions. Valid values are 0-9999 transactions.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 15

PROCESSING OPTIONS

The following fields define the processing options for AMEX.

ACQUIRER

A value indicating whether the institution acts as an acquirer of AMEX transactions. Valid values are:

Y = Yes, send transactions to AMEX.
N = No, do not send transactions to AMEX.

FIELD LENGTH: 1 alphabetic character
REQUIRED FIELD: Yes
DEFAULT VALUE: None

ISSUER

A value indicating whether the institution acts as an issuer of AMEX transactions. The AXCI ISO Interchange does not perform issuer processing. If two distinct station names are used on screen 1 and the station-ty parameter in the LCONF is set to 0 (normal processing), by setting this value to N, station 1 will be used as a second acquirer station. Valid values are:

Y = Do not use station 1 as an acquirer station.
N = Use station 1 as a second acquirer station.

FIELD LENGTH: 1 alphabetic character
REQUIRED FIELD: Yes
DEFAULT VALUE: None

PROCESSING MODE

The processing mode of the Interchange Interface process. The AXCI ISO Interface process does not use this field.

FIELD LENGTH: 1 alphanumeric character
REQUIRED FIELD: No
DEFAULT VALUE: None

AUTO SIGNON START

A value indicating whether the AXCI ISO Interface process is set to programmatically logon to AMEX at startup without operator intervention. Valid values are:

Y = Yes, logon.
N = No, do not logon.

FIELD LENGTH: 1 alphanumeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: None

MAXIMUM TIMEOUTS

The maximum number of consecutive timeouts allowed before network management measures are taken to determine the status of the link. Valid values are 0-9999.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 2

MAX SAF RETRY

The maximum number of times a transaction from the Store-and-Forward File (SAF) is transmitted to AMEX before it is dumped to the hard-copy log and deleted from the file. Valid values are 0-9999.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0

ACK TO SWITCH

A value indicating whether text-level acknowledgements are to be sent to AMEX.

FIELD LENGTH: 1 alphabetic character
REQUIRED FIELD: No
DEFAULT VALUE: None

ACK FROM SWITCH

A value indicating whether text-level acknowledgements are required from AMEX.

FIELD LENGTH: 1 alphabetic character
REQUIRED FIELD: No
DEFAULT VALUE: None

NETWORK MANAGEMENT MESSAGE ENABLED

A value indicating whether network management messages are enabled or disabled.

FIELD LENGTH: 1 alphanumeric character
REQUIRED FIELD: No
DEFAULT VALUE: None

TYPE OF INTERCHANGE REPORTS

A value indicating the types of reports the interchange requires.

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: No
DEFAULT VALUE: 0

ILF EXTRACT NUMBER

The number of ILFs to be extracted for host reporting. Valid values are 3-9. This field is used in conjunction with the RPT-EXTRACT field in the Extract Configuration File (ECF). For a description of the ECF, refer to the "BASE24 Base Files Maintenance Manual".

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: No
DEFAULT VALUE: 3

ICF SCREEN 7 SETTINGS

The Interface process does not use fields on ICF screen 7.

ICF SCREEN 8 SETTINGS

The Interface process does not use fields on ICF screen 8.

ICF SCREEN 9 SETTINGS

The following fields on ICF screen 9 must be set as indicated.

POS ICF DATA

The following fields define the transactions allowed for BASE24-pos.

TRANSACTIONS ALLOWED TO SWITCH

Values defining which transactions listed in the TRANSACTIONS ALLOWED TO SWITCH fields are allowed or disallowed. Valid values are:

Y = Yes, allowed.

N = No, not allowed.

The values in these fields are used for outbound transactions to determine whether BASE24 allows the transactions to be sent to AMEX. If a transaction is not allowed to be sent to AMEX, the AXCI ISO Interface process declines the transaction with a response code of 55 (ineligible transaction).

FIELD LENGTH:	1 alphabetic character
REQUIRED FIELD:	Yes
DEFAULT VALUE:	N

ICF SCREEN 10 SETTINGS

The following fields on ICF screen 10 must be set as indicated.

POS ICF DATA

The following fields contain information relevant to BASE24-pos.

AUTH PROCESS

The Authorization process that the AXCI ISO Interface process routes POS transaction requests to when it is servicing only one BASE24 node. If AMEX is servicing more than one logical network, the field should contain the word "GATEWAY" so the AXCI ISO Interface process knows to route the transaction through the Gateway process.

FIELD LENGTH:	1-16 alphanumeric characters
REQUIRED FIELD:	No
DEFAULT VALUE:	None

REFERRAL PHONE NUMBER

The phone number that the AXCI ISO Interface process places in all inbound requests if transaction is not involved in switch to switch routing. This number can be called for an authorization.

FIELD LENGTH: 1-18 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

RETAILER ID DEFAULT

The default retailer ID that the AXCI ISO Interface places in all internal requests.

FIELD LENGTH: 1-19 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

TIMEOUT ACTION

The action the AXCI ISO Interface process should take when a timer waiting for a response from the interchange expires before the response is received. Valid values are:

0 = Decline (default)
1 = Default action
2 = Alternate routing

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: 0

SETTLEMENT ENTITY

Used to determine the value of the draft capture field in the PSTM. Not used by AXCI ISO Interface process. Valid values are:

0 = No draft capture
1 = Draft capture

FIELD LENGTH: 1 alphanumeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: 0

TIMER LIMITS

The following fields set the time limits, in seconds, for BASE24-pos transactions.

STORE AND FORWARD

The time limit associated with the store-and-forward response from AMEX for POS transactions. Valid values are 0-9999 seconds.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 30

OUTBOUND

The time limit associated with transaction responses from AMEX for POS transactions. Valid values are 0-9999 seconds.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 15

INBOUND

The time limit associated with a transaction response from BASE24-pos. Valid values are 0-9999 seconds.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 15

COMPLETION

The time limit associated with completion messages in either direction for POS transactions.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: 60

COMPLETION ACK

The time limit associated with completion acknowledgement messages from an interchange. Not used by the AMEX Interface process.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: 60

ICF SCREEN 11 SETTINGS

The following fields on ICF screen 11 must be set as indicated.

POS PRODUCT DATA

The following fields contain information relevant to BASE24-pos.

DEFAULT PRE-AUTH AMOUNT

The default preauthorization amount to be used in an internal request when the external message does not send an amount. Not used by the AXCI ISO Interface process.

FIELD LENGTH: 1-5 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: 0

APPROVAL CODE LENGTH

The number of digits that an interchange response message can handle in an authorization identification response code.

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: No
DEFAULT VALUE: 6

PRE-AUTH HOLD INCREMENT

The time increment at which to hold the preauthorized funds. Valid values are:

0 = minutes (default)
1 = hours
2 = days

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: 0

PRE-AUTH HOLD TIME

The time length to hold the preauthorized funds. Valid values are 0-99. Not used by the AXCI ISO Interface process.

FIELD LENGTH: 1-2 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0

ALLOWED SERVICES

POS credit card service codes that identify the types of cards the AXCI Interchange allows. This field is used for both inbound and outbound POS messages. The BASE24-defined codes are:

AD = Administrative	AX = American Express
BD = Business deposit-only	C = Private label credit
CB = Carte Blanche	D = Demonstration
DC = Diners Club	M = MasterCard credit
MD = MasterCard debit	MM = MasterCard dual
P = Proprietary debit	V = VISA credit
VD = VISA debit	VV = VISA dual
** = all	

Institutions can also enter additional codes not listed above, as long as the first position of the code is an alphabetic character other than M or V (M and V are reserved for MasterCard and Visa).

ICF SCREEN 12 SETTINGS

The following fields on ICF screen 12 must be set as indicated.

1100 MESSAGE REASON CODE

This field contains a four-digit Message Reason Code, which is provided by American Express during certification. The code is used with 1100 request message submitted for processing by the merchant or third party processor.

FIELD LENGTH:	4 numeric character
REQUIRED FIELD:	Yes
DEFAULT VALUE:	0000

1420 MESSAGE REASON CODE

This field contains a four-digit Message Reason Code, which is provided by American Express during certification. The code is used with 1420 request message submitted for processing by the merchant or third party processor.

FIELD LENGTH:	4 numeric character
REQUIRED FIELD:	Yes
DEFAULT VALUE:	0000

1804 MESSAGE REASON CODE

This field contains a four-digit Message Reason Code, which is provided by American Express during certification. The code is used with 1804 request message submitted for processing by the merchant or third party processor.

FIELD LENGTH:	4 numeric character
REQUIRED FIELD:	Yes
DEFAULT VALUE:	0000

KEYF SETTINGS

Each interface process must have its own Key File (KEYF) record. The following describes how the KEYF must be configured for the AXCI ISO Interface.

The KEYF screens are not depicted; however, the fields are described as they actually appear on the Release 5.0 or above KEYF screens. Note that the AXCI ISO Interface requires settings on only the first screen, the remaining KEYF fields will contain the default values.

KEYF SCREEN 1 SETTINGS

The following fields on KEYF screen 1 must be set as indicated.

DPC/FIID

The FIID of the interchange using this KEYF record. The value in this field must match the value in the INTERCHANGE FIID field on ICF screen 1.

FIELD LENGTH: 1-4 alphanumeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: None

INTERFACE PROCESS

The name of the interface process associated with the interchange identified in the DPC/FIID field. The value in this field must match the value in the PROCESS field on ICF screen 1.

FIELD LENGTH: 1-16 alphanumeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: None

STF SETTINGS

The Switch Terminal-ID File (STF) must contain one record for each terminal that is required by AMEX to transmit an interchange-assigned identification rather than the BASE24 terminal identification.

STF screen 1 is not depicted; however, the fields are described as they actually appear on the screen. For actual screen depictions, refer to the "BASE24 Base Files Maintenance Manual".

STF SCREEN 1 SETTINGS

The following fields on STF screen 1 must be set as indicated.

SWITCH FIID

An identifier for the AMEX financial institution. The value entered in this field must be identical to the value entered in the INTERCHANGE FIID field on ICF screen 1.

FIELD LENGTH:	1-4 alphanumeric characters
REQUIRED FIELD:	Yes
DEFAULT VALUE:	None

MERCHANT ID

An identifier for the merchant. When BASE24-pos is used, this field is required. When BASE24-pos is not used, this field can be filled with asterisks or the terminal owner name from the TDF.

FIELD LENGTH:	1-19 alphanumeric characters
REQUIRED FIELD:	No
DEFAULT VALUE:	None

TERMINAL FIID

The FIID associated with the terminal.

FIELD LENGTH:	1-4 alphanumeric characters
REQUIRED FIELD:	No
DEFAULT VALUE:	None

TERMINAL ID

The BASE24 terminal identifier. The value entered in this field must be left-justified and blank-filled.

FIELD LENGTH:	1-16 alphanumeric characters
REQUIRED FIELD:	No
DEFAULT VALUE:	None

PROD ID

An identifier for the product with which the terminal is associated. Valid values are:

01 = ATM
02 = POS

FIELD LENGTH: 2 alphanumeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: None

SWITCH MERCHANT INFORMATION

The following fields contain information on the interchange merchant.

ID

The interchange merchant identifier to which the BASE24 merchant identifier should be translated. The value entered in this field should be left-justified and blank-filled.

FIELD LENGTH: 1-19 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

OFFSET

The position in the BASE24 merchant identifier that must be used as the first digit of the interchange merchant identifier.

FIELD LENGTH: 1-2 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0

LENGTH

The number of digits of the BASE24 merchant identifier that must be used as the interchange merchant identifier. The field should be allowed to default to 00 if this field and the OFFSET field are not used for translation purposes.

FIELD LENGTH: 1-2 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0

SWITCH TERMINAL INFORMATION

The following fields contain information about the interchange's terminal.

ID

The interchange's terminal identification or pseudo terminal number to which the BASE24 terminal identifier should be translated. The value entered in this field should be left-justified and blank-filled.

FIELD LENGTH: 1-19 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

OFFSET

The position in the BASE24 terminal identifier that must be used as the first digit of the interchange terminal identifier.

FIELD LENGTH: 1-2 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0

LENGTH

The number of digits of the BASE24 terminal identifier that must be used as the interchange terminal identifier. This field should be allowed to default to 00 if this field and the OFFSET field are not to be used for translation purposes.

FIELD LENGTH: 1-2 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0

ON PREMISE FLAG

A value indicating whether the terminal is at the same location as the institution that owns the terminal. Valid values are:

Y = On premise
N = Off premise

FIELD LENGTH: 1 alphanumeric
REQUIRED FIELD: No
DEFAULT VALUE: None

TOKEN FILE

If the AXCI ISO Interface is running in a BASE24 Release 5.0 or higher environment, the TOKEN FILE records for the AXCI ISO Interface can be entered (or updated). The token records are used to indicate which tokens to:

- a) Log to the Interchange Log File (ILF) and/or the POS Transaction Log File (PTLF).
- b) Extract from the Interchange Log File (ILF) and/or the POS Transaction Log File (PTLF).
- c) Send to the host.

The following tokens are currently supported by the AXCI ISO Interface:

- Alternate Merchant ID Token
- American Express Additional Data Token
- Amex Private Use Data Token
- Check Authorization Token
- EMV Request Data Token
- EMV Status Token
- EMV Response Data Token
- EMV Script Data Token
- Interchange Compliance Token
- Payment Service 2000 (PS2000) Token
- Point Of Service Data Token
- POS Data1 Token
- POS 50 (Release 5.0) Token
- POS 51 (Release 5.1) Token
- Switch Token
- Track 1 Token

Additional disk space may be required on the Tandem for the 5.0 or above BASE24 database depending on the number of tokens being logged to the files.

The Release 5.0 or above BASE24 user can indicate what tokens should be logged by the AXCI ISO Interface to its Interface Log File (ILF). The user can also indicate to the Release 5.0 or above BASE24-pos Auth modules which tokens should be logged to the POS Transaction Log File (PTLF).

When deciding whether or not to log a token to any Release 5.0 or above log file the user should consider what the length of the log record will be with the new token added. If logging a new token to a log file increases the length of a log file record so that fewer records can fit in the maximum Tandem data block, more data blocks (i.e. more disk space) will be required to hold the same number of records. This will increase the user's disk space requirements. The maximum Tandem data block is 4096 bytes, which is recommended for all BASE24 log files. Of the 4096 bytes, 22 bytes per block, and 2 bytes

per record are reserved for Tandem file system usage in the file type used for BASE24 log files.

Currently the AXCI ISO Interface can fit two ILF records per block, when taking into account Tandem file system usage, current data and space reserved for future enhancements. This leaves a maximum size of 2035 bytes per ILF record. For example adding the Interchange Compliance token to the ILF in Release 5.0 or above will not make the product ILF record greater than 2035 bytes, and therefore will not increase required disk space for the standard Release 5.0 or above AXCI ISO Interface.

However, if the Release 5.0 or above AXCI ISO Interface user has the Interface configured to support multiple tokens then the product ILF record could exceed the 2035 bytes when the total number of tokens are logged to the ILF.

Also, if the Release 5.0 or above AXCI ISO Interface user is currently logging any RPQ tokens to the ILF or has modified the ILF in any other manner, the user should determine if logging the desired tokens will put the length of their AXCI ISO Interface ILF records past 2035 bytes. If only one token is logged to the ILF, the ILF will increase by the length of the token, plus an additional 6 bytes due to a Header token also being added. All subsequent tokens will only increase the ILF record length by the length of the respective token.

It is not necessary to log the Interchange Compliance token to the ILF because the information is already present in the external message section. However, if the user is settling using the ILF and would rather retrieve the data from the token, they may do so.

Neither is it necessary to log the PS2000 token to the ILF for online transaction processing, as the AXCI ISO Interface will recreate any necessary PS2000 data from messages received from Visa through American Express from the PS2000 data logged in the ILF as part of the AXCI external message section. However, the user may want to log it for other purposes such as reporting and not have to search the external message section of the ILF for the data.

The user should do the same analysis on their 5.0 or above BASE24-pos log file (PTLF) as done for the AXCI ISO Interface ILF record size. If the extra bytes added to the PTLF due to tokens increase the size of the PTLF records enough to decrease the number of records that will fit in Tandem's maximum data block, the user may need more disk space.

Additional disk space is not required on the Tandem for the Release 3.4/4.0 BASE24 database because 200 bytes of user data is always logged to the ILF and the PTLF.

Section 3 STARTUP AND CONTROL

INTRODUCTION

This section provides information on the configuration of the AXCI ISO Interface process. It also describes process initialization, network control commands, and timers.

PROCESS CONFIGURATION

The following paragraphs explain how the AXCI ISO Interface process is configured. They provide information about the startup logic of the AEGN ISO Interface process and the method in which the AXCI ISO Interface process logs messages.

STARTUP

Because of the length of time required for initialization, the AXCI ISO Interface process should be assigned a startup logic of AUTOMATIC in the network environment. Automatic means the AXCI ISO Interface process is started automatically when the XPNET node is initiated. This ensures that the AXCI ISO Interface process is ready when it receives its first transaction.

EVENT MESSAGES

The AXCI ISO Interface process sends all of its event messages to the logging facility using a standard routing code of zero.

INITIALIZATION

The following steps are taken by the AXCI ISO Interface process when it is started.

1. Initializes interprocess communication with the network.
2. Reads the name of its Logical Network Configuration File (LCONF). The startup message from the XPNET process contains the name of the LCONF. This name allows the AXCI ISO Interface process to access the LCONF for the assigns and params it requires for processing.
3. Opens the LCONF, reads the required assigns and params, and closes the LCONF. The process will abend if there is a problem reading the LCONF assigns or params or if a required assign or param is missing. For a list of the required LCONF assigns and params and their descriptions, refer to section 2.
4. Reads the Log Message Configuration File (LMCF) for its modified log messages. The LMCF contains a record for each modified log message in the system. The AXCI ISO Interface process opens the LMCF, retrieves its modified log messages, and closes the LMCF. Whenever the AXCI ISO Interface process generates a log message, it checks the LMCF records in memory to determine whether the message requires modifications.
5. Opens the Interchange Configuration File (ICF). If an error occurs, the AXCI ISO Interface process logs an error message and abends.
6. Reads its own ICF record using its process name.
7. Opens the Key File (KEYF). If an error occurs, the AXCI ISO Interface process logs an error message and abends.
8. Using its process name, reads its own KEYF record, assigns process control table (PCT) values, and closes the KEYF.
9. Finishes building the process control table with information from the ICF. The PCT contains configuration and control information for the process. This information includes all of the controls required to process transactions.
10. Determines the posting dates for BASE24-pos and AMEX by comparing the current time to the BASE24-atm and BASE24-pos time taken from the LCONF.
11. Initializes the extended memory required for this process.
12. Opens the Store-and-Forward File (SAF). If an error occurs, the AXCI ISO Interface process logs an error message and abends.
13. Opens the current and previous day's Interchange Log File (ILF). If an error occurs in opening the current ILF, the AXCI ISO Interface process logs an error message and abends.

14. Initializes the security devices.
15. Sends a logon message to AMEX if the automatic logon option was selected or an advice logon message if the advice logon option was selected.
16. Closes the ICF.
17. Reads the Token File into extended memory if tokens from the PSTM are not to be logged to the ILF.
18. If reports are required, starts a timer set to expire when the reports are to be run.
19. Set a cutover timer.
20. If the automatic logon option was selected, sends a logon message to AMEX.
21. Waits for a message from AMEX or a BASE24 process.

NETWORK CONTROL COMMANDS

The commands described below can be sent to the AXCI ISO Interface process. Each description includes the command format and a description of what happens when the message is received.

Network control commands can be issued from the Network Control Supervisor's (NCS) screen or from the NCPcom prompt.

The commands are entered using the following syntax:

DELIVER PROCESS (process name), "(Message Format)"

DEBUG

Message Format: DEBUG

Description: Places the interface process in DEBUG. This command should only be used in a TEST environment due to negative performance impacts.

DUMP

Message Format: DUMP

Description: Initiates a programmatic dump of the interface process. This command should only be used in a TEST environment or upon the request of ACI.

LOGOFF

Message Format: LOGOFF

LOGOFF <station>

Description: Initiates a logoff sequence. This command causes the AXCI ISO Interface process to mark the station(s) logged off. A logoff message is not sent to AMEX. If the interface is processing with two acquirer stations, both will be logged off.

LOGON

Message Format: LOGON

LOGON <station>

Description: Initiates a logon sequence. This command causes the AXCI ISO Interface process to send a logon message to the station(s) assigned on screen 1 of the ICF.

If using two acquirer stations, the AXCI ISO Interface process sends a logon message to AMEX on both stations.

PERFORMANCE

Message Format: PERFORMANCE

Description: Requests a display of the performance statistics kept in memory by the interface process. These performance statistics are written to the log via a series of log messages generated by the process. The following is an example of a performance display:

```
*** <POS> Performance Statistics Summary ***
Minutes in period: <#> <Outbound/Inbound> rqst limit: <#>
      REQUESTS - TIMEOUTS - PERCENT - AVERAGE
PERIOD <#>      <####>      <####>      <####>%    <##.##>
PERIOD <#>      <####>      <####>      <####>%    <##.##>
PERIOD <#>      <####>      <####>      <####>%    <##.##>
PERIOD <#>      <####>      <####>      <####>%    <##.##>
PERIOD <#>      <####>      <####>      <####>%    <##.##>
```

STATUS

Message Format: STATUS
Description: Requests the current status of the interface process. The status information is written to the log via a series of log messages generated by the process. The following is an example of the status information:

```
BASE24-pos DATE:  <YYMMDD>
SWITCH DATE:      <YYMMDD>
<#> outbound requests pending of <#> maximum requests
<#> inbound requests pending of <#> maximum requests
<#> timers active of <#> maximum timers
FILE: <ILF filename>, <#> PERCENT FULL
FILE: <SAF filename>, <#> RECORDS, <#> PERCENT FULL
SAF cycle in progress
FILE: <SDF filename>, <#> RECORDS, <#> PERCENT FULL
RSAF cycle in progress
TRACE (%B <trace bit values>)
Station <station> is <UP/DOWN> and LOGGED <ON/OFF>
      <as ISSUER/ACQUIRER>
Station <station> has <#> timeouts of <#> maximum timeouts
```

TRACE

Message Format: TRACE ON
TRACE OFF
Description: The interface process contains trace log messages that are optionally logged. The trace log messages should be requested only in a TEST environment due to negative performance impacts.

If "ON" is specified in the message, the interface starts the trace function and sends log messages to the current logging device.

If "OFF" is specified in the message, the interface stops the trace function.

CUTOVER OR "9502"

Message Format: 9502 YYMMDD
Description: This causes the AXCI ISO Interface process to increment the date sent in the message, YYMMDD, by one and set the internal BASE24 dates to the incremented value. This will only be done if the date in the message, YYMMDD, is greater than or equal to the current internal BASE24 dates.

WARMBOOT OR "9501"

Message Format: WARMBOOT
9501

Description: Warmboots the interface process. A message is logged to indicate whether the warmboot completes or aborts.

TIMERS

This section describes each timer used by the AXCI ISO Interface process. Each timer description includes the timer's function, and how and when the timer is set.

OUTBOUND REQUESTS TIMER

An outbound request timer is set each time a 1100 or 1420 message is sent to AMEX. The number of seconds used for this timer is taken from the OUTBOUND field on ICF screen 8 for BASE24-atm transactions or ICF screen 10 for BASE24-pos transactions. The outbound request timer indicates whether a transaction sent to AMEX has timed out.

If the timer expires before a response is received from AMEX for the 1100 or 1420 message, the request is denied and the appropriate performance and timeout counters are increased. If the expiration of this timer causes the maximum allowed timeouts to be reached, the link to AMEX is marked down.

INBOUND REQUESTS TIMER

An inbound request timer is set each time an 0200 message is sent to a BASE24 process for authorization. The number of seconds used for this timer is taken from the INBOUND field on ICF screen 10 for BASE24-pos transactions. The inbound request timer is used to determine that an external transaction has timed out.

If the timer expires before a response is received from the authorizing process, the request is denied. The appropriate performance and timeout counters are incremented.

STORE-AND-FORWARD TIMER

A store-and-forward timer is set each time a store-and-forward message is sent to AMEX. The number of seconds used for this timer is taken from the STORE AND FORWARD field on ICF screen 10 for BASE24-pos transactions. The store-and-forward timer indicates whether a Store-and-Forward File (SAF) record sent to AMEX has timed out.

If the timer expires before an acknowledgement is received from AMEX, the store-and-forward timeout counter is increased. If the store-and-forward timeout counter is equal to the maximum number of retries set in the MAX SAF RETRY field on ICF screen 3, the record is written to the log and deleted from the SAF. If the MAX SAF RETRY field is set to 0, the store-and-forward message is sent indefinitely until it is successfully processed.

CUTOVER TIMER

A cutover timer is set to handle cutting over the switch business date and creating an ILF for that date, if the business date has not already been changed.

NETWORK MANAGEMENT TIMER

A network management timer is set each time an 1804 message is sent to a station. The number of seconds used for this timer is taken from

the `TIMER LIMITS NETWORK MANAGEMENT` field on ICF screen 3. The network management timer is used to monitor the communications between BASE24 and AMEX.

If the network management timer expires before a response is received from AMEX, an extended network management timer is set.

EXTENDED NETWORK MANAGEMENT TIMER

An extended network management timer is set each time a network management timer expires without a response being received from AMEX. The number of seconds used for this timer is taken from the `TIMER LIMITS EXTENDED NETWORK` field on ICF screen 3.

The network management message is repeated at the interval specified by the extended network management timer until an acknowledgement is received from AMEX or the communications line is marked up.

WAIT FOR TRAFFIC TIMER

A wait for traffic timer is set each time a station is marked up and each time a previously set wait for traffic timer expires. The number of seconds used for this timer is taken from the `TIMER LIMITS WAIT FOR TRAFFIC` field on ICF screen 3.

If the timer expires and message traffic has occurred on the line, a new wait for traffic timer is started.

If the timer expires and no message traffic has occurred on the line, a network management logon message is sent. The line is marked up or down, depending on the results of the logon message.

REPORT STARTUP TIMER

The report startup timer is used to start the SETL01 and STAT09 report programs. It is set at initialization and at interchange cutover. The value for this timer is taken from the `SW-RPT-STARTUP` param in the `LCONF`.

When the report startup timer expires, the obey file containing the commands to run the report programs are executed.

PERFORMANCE TIMER

A performance timer is set at the start of each new performance interval. The number of minutes for this timer is taken from the `TIMER LIMITS PERFORMANCE PERIOD` field on ICF screen 3. The performance timer triggers the end of one performance period and the beginning of the next.

When the performance timer expires, the AXCI ISO Interface process:

1. Moves the performance statistics for interval four to interval five.
2. Moves the performance statistics for interval three to interval four.

3. Moves the performance statistics for interval two to interval three.
4. Moves the performance statistics for interval one to interval two.
5. Initializes the performance statistics for interval one to zero.
6. Starts a new performance timer.

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Section 4 TRANSACTION HANDLING

INTRODUCTION

This section describes how the AXCI ISO Interface process handles various types of transactions. It describes the basic flow of transaction processing and provides transaction flow diagrams illustrating how messages to and from AMEX are handled.

The following transaction flows are described in this section.

- Requests
- Reversals
- Timeouts
- Store-and-Forward
- Echo Test

NOTE: References in this section to transactions sent to AMEX for authorization implies that AMEX is sending the transaction to the appropriate authorizing entity (i.e., the AMEX Interchange does not perform authorization).

LOGGING TO THE ILF AND SAF

Requests and reversals messages handled by the AXCI ISO Interface process are logged to the Interchange Log File (ILF). Reversals are also logged to the Store-and-Forward File (SAF).

TRACE NUMBERS

When the AXCI ISO Interface process receives an outbound transaction request, it assigns a trace number to the transaction. The trace number is sent to AMEX in Field 11 of the Standard External Message (SEM). Trace numbers are used to uniquely identify a transaction, and a new trace number is assigned to each new request. Reversal transactions contain the trace number of the original transaction in Field 56 (Original Data Elements) of the message. Trace numbers begin at 00000 and are incremented by 1 until 32767 is reached. At this point, the trace number begins again at 00000.

Trace numbers must be unique, so the AXCI ISO Interface process must keep track of trace numbers it is assigning. The AXCI ISO Interface process determines the current trace number during initialization by reading the trace record from the current or previous day's ILF. This trace record is updated after a set number of transactions have been processed. The number of transactions processed before updating the trace record is set by a parameter in the Logical Network Configuration File (LCONF). Using this technique, the AXCI ISO Interface process can avoid creating duplicate trace numbers even in situations where the process is stopped and restarted.

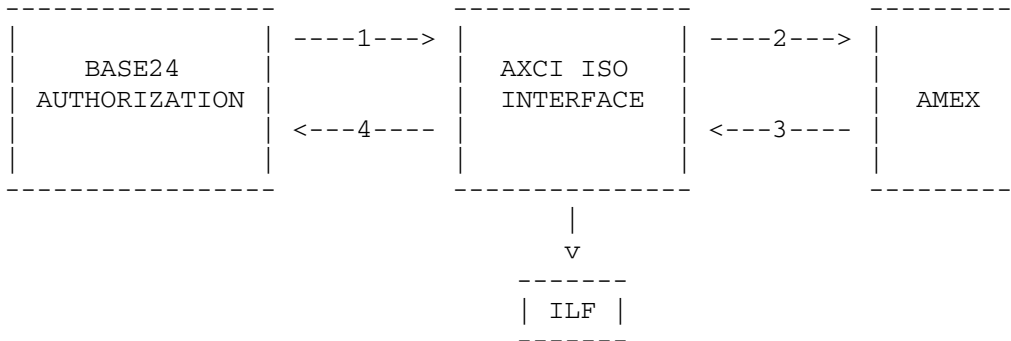
MESSAGE FLOWS FROM BASE24 TO AMEX

This subsection describes the message flows for the following transactions outbound from BASE24 to AMEX.

- Requests
- Reversals
- Timeouts
- Echo Test
- Store-and-Forward

REQUEST MESSAGE TO AMEX

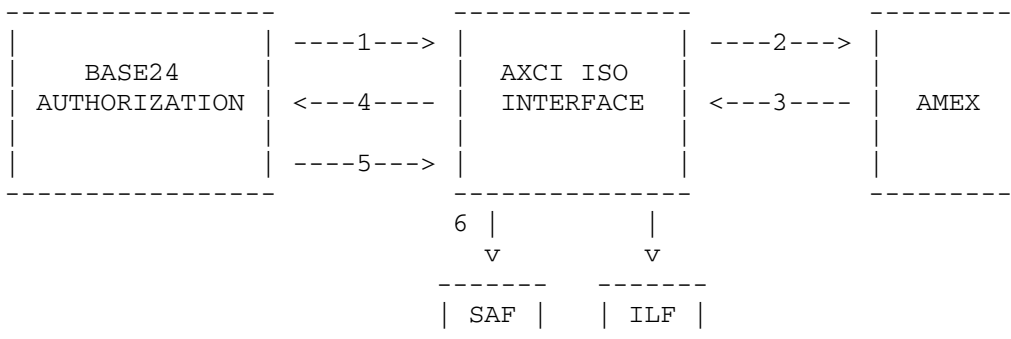
The diagram below illustrates the message routing that takes place when a transaction is initiated by the BASE24 network and must be sent to AMEX for authorization. The diagram steps are described below.



1. A BASE24 Authorization process sends the request initiated at a POS to the AXCI ISO Interface process in the 0200 internal message format.
2. The AXCI ISO Interface process reformats the message into the 1100 external message format and sends the external message to AMEX for authorization. An outbound request timer is set to wait for a response from AMEX.
3. AMEX routes the transaction to the appropriate authorizer and receives the subsequent response. The response is formatted into the 1110 external message format and sent to the AXCI ISO Interface process.
4. The AXCI ISO Interface process deletes the outbound request timer and formats a 0210 internal response message. The response is returned to the original BASE24 Authorization process and completed at the POS. The transaction is logged to the Interchange Log File (ILF).

REVERSAL TO AMEX

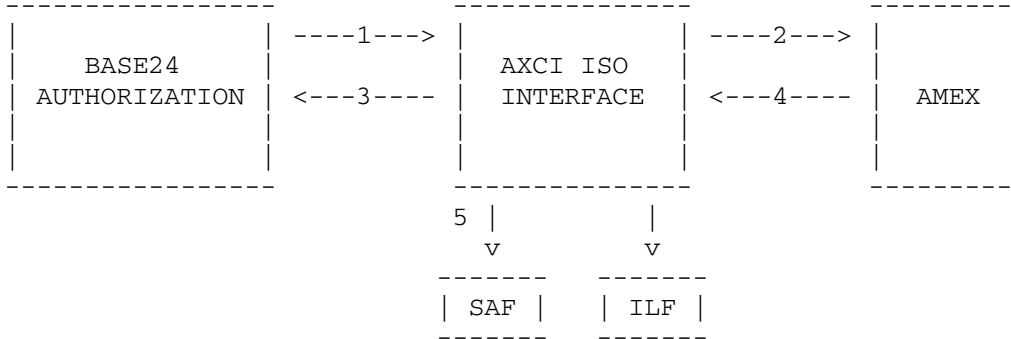
When a POS connected to the BASE24 network fails to complete a transaction as authorized, a reversal message must be sent to AMEX. The diagram below illustrates message routing when a reversal is sent to AMEX. The diagram steps are described below.



1. A BASE24 Authorization process sends the request initiated at a POS to the AXCI ISO Interface process in the 0200 internal message format.
2. The AXCI ISO Interface process reformats the message into the 1100 external message format and sends the external message to AMEX for authorization. An outbound request timer is set to wait for a response from AMEX.
3. AMEX routes the transaction to the appropriate authorizer and receives the subsequent response. The response is formatted into the 1110 external message format and sent to the AXCI ISO Interface process.
4. The AXCI ISO Interface process deletes the outbound request timer and formats a 0210 internal response message. The response is returned to the original BASE24 Authorization process and completed at the POS. The transaction is logged to the Interchange Log File (ILF).
5. The BASE24 Authorization process returns a 0420 message to the AXCI ISO Interface process, indicating the transaction failed to complete as authorized. The AXCI ISO Interface process locates the ILF record for the original transaction using the primary key it assigned to the transaction. The AXCI ISO Interface process then updates the ILF record with the reversal information.
6. The AXCI ISO Interface process formats a 1420 external message. The external message is logged to the Store-and-Forward File (SAF) and sent to AMEX during SAF processing.

TIMEOUTS OF REQUESTS TO AMEX

Each time the AXCI ISO Interface process sends an 1100 financial transaction request to AMEX for authorization, the AXCI ISO Interface process sets an outbound request timer. If the outbound request timer expires before a response is received from AMEX, the transaction times out. The diagram below illustrates message routing when a financial transaction times out. The steps in the diagram are described below.



1. A BASE24 Authorization process sends the request initiated at a POS to the AXCI ISO Interface process in the 0200 internal message format.
2. The AXCI ISO Interface process reformats the message into the 1100 external message format and sends the external message to AMEX for authorization. An outbound request timer is set to wait for a response from AMEX.
3. When the timer expires before a response is received from AMEX, the AXCI ISO Interface process formats a 0210 internal response message denying the transaction. The response is returned to the BASE24 Authorization process. The transaction is logged to Interchange Log File (ILF).
4. AMEX sends a 0210 late response to the AXCI ISO Interface process. If the transaction was denied by AMEX, the AXCI ISO Interface process simply drops the late response.
5. If the transaction has been approved by AMEX, the AXCI ISO Interface process formats a 1420 external reversal message and adds it to the Store-and-Forward File (SAF) to send to AMEX during SAF processing.

STORE-AND-FORWARD MESSAGES TO AMEX

Non-interactive messages intended for AMEX are stored in the Store-and-Forward File (SAF) and sent to AMEX during the SAF processing cycle. After the processing of a message is completed, the AXCI ISO Interface process checks to determine whether a SAF record should be sent. If the link to AMEX is up, the SAF contains records, and a SAF record is not in the process of being sent to AMEX, the AXCI ISO Interface process retrieves the first record from the SAF and sends it to AMEX. Only one message from the SAF is sent at a time. The general processing flow for SAF messages is the same as for normal messages. To AMEX, the messages appear to be happening as they are sent. Therefore, AMEX cannot distinguish them from normal messages. The AXCI ISO Interface process uses different timers that identify the messages as SAF records. The following message type can be logged to the SAF:

- 0420 Reversal messages

AMEX sends acknowledgements of these messages to the AXCI ISO Interface process. A 1430 message is an acknowledgement to the reversal message. When the AXCI ISO Interface process receives the acknowledgement, the SAF timer is deleted and the SAF record is deleted from the SAF. The acknowledgement message from AMEX is dropped.

NETWORK MANAGEMENT MESSAGE FLOWS

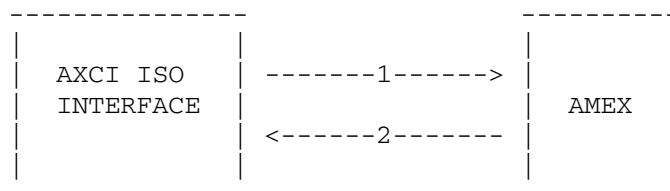
Network management messages are used to manage the operational status of the communications lines between BASE24 and AMEX. The AXCI ISO Interface process supports the following types of network management messages.

- Logon messages
- Echo test messages

These messages are sent as 1804 messages and require 1814 messages in response. This section describes the message flows that occur when network management messages are sent.

LOGON/ECHO TEST INITIATED BY BASE24

The diagram below illustrates the message flow between the AXCI ISO Interface process and AMEX when BASE24 initiates a logon or echo test message. The steps in the diagram are described below.



1. To initiate a logon, the AXCI ISO Interface process sends an 1804 logon request with a function code of 831 to AMEX. This message is used to establish communications between BASE24 and AMEX. When a wait-for-traffic timer expires two consecutive times with no line traffic in between, the AXCI ISO Interface process sends an 1804 echo test message with a function code of 831 to AMEX. This message is used to ensure the link is still up between the AXCI ISO Interface process and AMEX.
2. AMEX responds with an 1814 message. If the action code is set to 800, the response indicates the logon/echo test has been accepted. If the action code is not set to 800, the response indicates the logon/echo test was rejected.

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Section 5 CUTOVER AND REPORTING

INTRODUCTION

This section describes the AXCI ISO Interface process's cutover processing and report generation.

CUTOVER PROCESSING

Within the BASE24 system, interchange cutover processing is a transaction management and reporting function that produces reports for each business day. These reports are provided to aid in reconciling the BASE24 network with AMEX. AMEX initiates cutover seven days a week.

REPORTING

The AXCI ISO Interface process is responsible for starting the interchange reports. It does so by creating an obey file to start the report programs and then running the obey file at the appropriate time. If the report related assigns and params are not set in the LCONF, the reports will not be generated.

Using the time specified in the LCONF param SW-RPT-STARTUP, the AXCI ISO Interface process sets a timer to expire when the reports are to be run. When the timer expires, the AXCI ISO Interface process creates and runs an obey file containing the commands necessary to run the report programs from Tandem's TACL.

Under normal circumstances, the report programs start and stop without operator intervention. However, if the AXCI ISO Interface process does not start the programs, they can be started manually from a TACL prompt by running the report obey file created by the AXCI ISO Interface process. If the report programs abend or fail, they can be restarted in the same way.

REPORT OVERVIEW

The AXCI ISO Interface process produces two reports of interchange transaction activity. These reports are the SETL01 Clearings Statement Report and the STAT09 Transaction Detail Report. These reports can be used in conjunction with BASE24-pos reports to reconcile the differences between the BASE24 processing day and the AMEX processing day.

This section provides an overview of the information contained in the interchange reports. For more detailed information and sample report formats, see the "BASE24-pos Settlement and Reporting Manual".

SETL01 CLEARINGS STATEMENT REPORT

The SETL01 Clearings Statement Report reflects the debit, credit, and net position of AMEX to BASE24 in the customer's currency. This report contains the totals of all approved transactions logged to the Interchange Log File (ILF). The first section of the report gives transaction totals for the BASE24 processing day.

Following the BASE24 totals, a section exists for transactions that occurred during the BASE24 processing day but not during the AMEX processing day. These transactions are subtracted from the transaction totals for the BASE24 processing day. This section of the report provides the net amount of transactions and the dollar difference between transactions occurring during the BASE24 processing day but not during the AMEX processing day.

Finally, a section exists showing transactions that occurred during the AMEX processing day but not during the BASE24 processing day. These transactions are added to the transaction totals for the BASE24 processing day. This section of the report provides the net amount of transactions and the dollar difference between transactions occurring during the AMEX processing day but not during the BASE24 processing day.

STAT09 TRANSACTION DETAIL REPORT

The STAT09 Transaction Detail Report provides a complete listing of all AMEX activity. The report is based on the transactions logged to the current day's ILF.

This report contains one detail item for each transaction processed by the AXCI ISO Interface process during its processing day. The information provided includes the cardholder number, transaction type, transaction time, posting date, and transaction amount.

STARTING REPORTS

The AXCI ISO Interface process is responsible for starting the interchange reports. It does so by creating an obey file to start the report programs and then running the obey file at the appropriate time. If the report related assigns and params are not set in the LCONF, the reports will not be generated.

Using the time specified in the LCONF param SW-RPT-STARTUP, the AXCI ISO Interface process sets a timer to expire when the reports are to be run. When the timer expires, the AXCI ISO Interface process creates and runs an obey file containing the commands necessary to run the report programs from Tandem's TACL.

Under normal circumstances, the report programs start and stop without operator intervention. However, if the AXCI ISO Interface process does not start the programs, they can be started manually from a TACL prompt by running the report obey file created by the AXCI ISO Interface process. If the report programsabend or fail, they can be restarted in the same way.

Appendix A RDT File

Introduction

This information is an example of the compile commands used by Tandem's SNAX/HLS product. The SNAX/HLS operating environment presumes that Resource Definition Table (RDT) exists. The RDT accessed by SNAX/HLS is created by compiling an edit file. The commands shown below can be used as input to the SNAX/HLS compiler to create the RDTDATA file. For further information regarding the RDTDATA, RDTTEXT and HLSRDT, see the Tandem "SNAX/HLS Configuration and Control Manual".

```
reset BIND
!Bind BAMEXLU1 BIND IMAGE FOR AMEX DIALUP TESTING
set bind bind-type          NONNEGOTIABLE
set bind fm-profile         4
set bind ts-profile         4
set bind pri-chain-use      MULTIPLE-RU
set bind pri-rq-mode        IMMEDIATE
set bind pri-chain-protocol ANY
set bind pri-two-phase-commit FALSE
set bind pri-compression-ind FALSE
set bind pri-eb-send        TRUE
set bind sec-chain-use      MULTIPLE-RU
set bind sec-rq-mode        IMMEDIATE
set bind sec-chain-protocol ANY
set bind sec-two-phase-commit FALSE
set bind sec-compression-ind FALSE
set bind sec-eb-send        TRUE
set bind no-segments        FALSE
set bind fm-header-use      FALSE
set bind brkt-reset         BETB
set bind brkt-term          CONDITIONAL
set bind alt-code-use       FALSE
set bind seqno-use          FALSE
set bind bis-sent           FALSE
set bind bind-queuing       FALSE
set bind send-recv          FDUX
set bind recovery-mode      LOSER
set bind contention-winner  SECONDARY
set bind hdx-ff-reset-sender SECONDARY
set bind pri-send-pacing-size 0
set bind sec-send-pacing-size 0
set bind sec-max-ru         256
set bind pri-recv-pacing-size 0
set bind sec-recv-pacing-size 0
set bind pri-max-ru         256
set bind lu-session-type    0
set bind ps-usage           000000000000000000000000
```

```
set bind user-data          0
set bind plu-name          STRP
add BIND                    BAMEXLU1
```

```
!-----
```

```
reset PROFILE
! Profile PAMEXLU1 PROFILE FOR AMEX DIALUP TESTING
set profile bind-name      BAMEXLU1
set profile open-mode      ACQUIRE
set profile hs-polarity    PRIMARY
set profile want-sscp-text FALSE
set profile response-type  3
set profile pipeline-lu-ind FALSE
set profile translate      FALSE
set profile lmo            FALSE
set profile max-requests   2
set profile want-bind-data NONE
set profile user-responses FALSE
set profile want-flow-control FALSE
set profile encode-eri     FALSE
set profile single-fmh     FALSE
set profile omit-modem-wait FALSE
add PROFILE                PAMEXLU1
```

The above information is for BASE24's support of the Primary Logical Unit (PLU) protocol. To configure the RDT File for BASE24's support of the Secondary Logical Unit (SLU-P) protocol the following changes need to be made:

Delete the param "bind-name" in the settings for the profile.

Change the profile params "hs-polarity" to SECONDARY and "open-mode" to ACCEPT.

The section that sets and adds the bind parameters should be deleted.

CREATING AN RDT FILE

An RDTDATA file can be created by the following set of commands.

1. Edit the source file using the following command:

```
> EDIT RDTTEXT  
  (Perform changes)  
  
*EXIT
```

2. Compile the RDTTEXT file using the following command:

```
> RUN HLSRDT / in RDTTEXT, out $S.#LP / RDTDATA
```

OPERATING SNAX/HLS

The following set of commands is used to start the new HLSOBJ. For further details, see the section on operating SNAX/HLS of the "SNAX/HLS Configuration and Control Manual" for more details.

```
> stop $hlsx
> hlsobj/in rdtdata, out $sudo, nowait, name $hlsx, pri 140/
  abend^on^error on, max^lus 32, initial^trace 15
```

where the PPD name \$hlsx will be used in the port name for the line declaration.

CFI OBEY FILE

The following is a list of Communication Management Interface (CFI) commands to start the SNA sessions. For further information on CFI, see Tandem's "Communication Management Interface Operator Reference Manual"

```
allow all errors
assume line $snap1
abort lu under
abort pu under
abort
delete lu under
delete pu under
delete
add line, characteraset ascii
add pu $snap1.#pu1,type(13,2),recsize 265,maxlus 2,address %B0, &
    window 7
add lu $snap1.#lu1,type(14,2),puname #pu1,recsize 265,protocol &
    snalu, address 1, characteraset ascii
start
start pu under
start lu under
```

Note - The value of addresses mentioned above were set to match the American Express Test System. These might need to be changed depending upon the configuration.

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Appendix B NEFS DECLARATIONS

Introduction

The following information gives example Device, Station and Line declarations that can be used when communicating with Amex utilizing an X.25 connection using the HOST ETE protocol. These definitions should be declared in the users NEFS. For further information see the "BASE24 XPNET 2.0 Network Environment Source file manual".

```
USERDATA:      !USERDATA may or may not need to be configured for the
                !station
                END^TO^END^USERDATA (4 BYTES);
```

```
PROCESS  P1A^AXCI1 = 48:
          OBJECT "$SWITCH.SW53AXCI.AXCI",
          LIBRARY "$SYSTEM.SPANNET.SKELB",
          QUEUE ALERT THRESHOLD 70,
          QUEUE LIMIT 150,
          PPD NAME IS "T1B1",
          LOGICAL NETWORK IS AXCI,
          CLASS "SWITCH",
          BOJ,
          PRIORITY 118,
          CPU 0 & 1;
```

```
DEVICE  AXCI1^SWITCH^X25 = 27:
          DEVICE TYPE      29,
          RECEIVE          4000,
          TRANSMIT         1000,
          TRANSMISSION NUMBER LENGTH 0,
          NORMAL SELECT;
```

```
STATION  S1AAXCI1 = 34:
          DEVICE AXCI1^SWITCH^X25,
          LOGICAL NETWORK IS TES1,
          DESTINATION PROCESS P1A^AXCI1,
          QUEUE ALERT THRESHOLD 10,
          CLASS "SWITCH",
          QUEUE LIMIT 100,
          REINSTATE EVERY 30 SECONDS,
          END^TO^END^USERDATA [0,0,0,0],
          RETRY 1;
```

```
LINE  L1A^AXCI1 = 30:
          DEVICE ID "$X25XA1.#PU01",
          X25,
          END TO END PROTOCOL 9,
          RECALL DELAY 02,
          REMOTE " ",
          FULL DUPLEX,
          PVC,
          ATIMER 10,
```

```
BTIMER 20,  
STATIONS ( S1AAXCI1 );
```

CMi OBEY FILE

The following is a list of Communication Management Interface (CMI) commands to start the X.25 sessions. For further information on CMI, see Tandem manual "Communication Management Interface Operator Reference Manual".

The customers system management group should be responsible for correctly configuring the X.25 ports. Any sample configuration we provide may not cover all of the available options the customer may wish. Here is a sample configuration and it may not include how the customer may eventually implement its X.25 configuration.

The user will also have to type the following commands under CMI. (D customers will be using SCF. All the CMI commands will work under SCF but the only change will be in the 'add su' statement. The 'type' will change to 'devtype' in the 'su' statement.)

```
assume line $<ldev name>  
abort, sub all  
alter, pvcrange (1,2), svcrange (3,255), clockspeed clock19200, pvc 1  
add    su #<port name>, recsize 4095, protocol ptp, type(9,0)  
start,sub all  
status,sub all
```

SYSGEN LINE CONFIGURATION

The following is a list of SYSGEN Line Configuration commands needed to start the X.25 sessions. For further information, see Tandem manual "System Generation Manual for X25AM".

```
DELETE  $X25XA1  
SET MACRO X25AM  
SET PATHNAME PSYNC0  
SET SUBTYPE 63  
SET RSIZE 128  
SET LIU 1  
SET CIRCUITS 255  
SET PACKETSIZE 256  
SET L2WINDOW 7  
SET L2RETRIES 20  
SET L2TIMEOUT 300  
SET L2IDLETIMEOUT 500  
SET L3WINDOW 2  
SET IDLECMDRR  
SET L2DISC  
SET NOEXTFORMAT  
SET RS232  
SET DTE
```

Appendix A

```
SET SYNC3 3
SET STARTDOWN
SET AUTOLOAD
ADD      $X24XA1
START    $X25XA1
```

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